

## **LABORATORY ANIMALS**

Laboratory Animals are generally defined as any vertebrate animal produced for or used in research, testing or teaching. Animal use is defined as the proper care, use and humane treatment of laboratory animals produced for or used in research, testing or teaching.

**Ethics and Animal Use:** The decision to use animals in research requires critical thought, judgement and analysis. Using animals in research is a privilege granted by the society to the research community with the expectation that such will provide either significant new knowledge or lead to improvement in human and/or animal well being. It is a trust that mandates responsible and humane care and use of these animals.

**The Three Rs:** these represent the practical method for implementation of the principles guiding the ethical use of animals in Research. The Three Rs have become an internationally accepted approach for researchers to apply when deciding to use animals in research and in designing humane animal research studies. They are Replacement, Refinement and Reduction.

**Replacement** refers to methods that avoid using animals. The term includes absolute replacements (i.e., replacing animals with inanimate systems such as computer programs) as well as relative replacements (i.e., replacing animals such as vertebrates with animals that are lower on the phylogenetic scale).

**Refinement** refers to modifications of husbandry or experimental procedures to enhance animal well-being and minimize or eliminate pain and distress. While institutions and investigators should take all reasonable measures to eliminate pain and distress through refinement, IACUCs (Institutional Animal Care and Use Committees) should understand that with some types of studies there may be either unforeseen or intended experimental outcomes that produce pain. These outcomes may or may not be eliminated based on the goals of the study.

**Reduction** involves strategies for obtaining comparable levels of information from the use of fewer animals or for maximizing the information obtained from a given number of animals (without increasing pain or distress) so that in the long run fewer animals are needed to acquire the same scientific information. This approach relies on an analysis of experimental design, applications of newer technologies, the use of appropriate statistical methods, and control of environmentally related variability in animal housing and study areas

Refinement and reduction goals should be balanced on a case-by-case basis. Principal investigators are strongly discouraged from advocating animal reuse as a reduction strategy, and reduction should not be a rationale for reusing an animal or animals that have already undergone experimental procedures especially if the well-being of the animals would be compromised. Studies that may result in severe or chronic pain or significant alterations in the animals' ability to maintain normal physiology, or adequately respond to stressors, should include descriptions of appropriate humane endpoints or provide science-based justification for not using a particular, commonly accepted humane endpoint. Veterinary consultation must occur when pain or distress

is beyond the level anticipated in the protocol description or when interventional control is not possible.

## **Types and Use of Laboratory Animals**

### **Advantages of Animals**

1. Isolation of microorganisms.
2. Identification of microorganisms.
3. Preparation of immune sera/therapeutic sera diagnostic sera/vaccines (antigens).

### **Disadvantages of Animals**

1. Ethical issues regarding uses of animal. Required to take permission from the animal ethical committee of institute.
2. Time consuming about 6–9 months.
3. May not give desired results due to interference by animal immunity.

Laboratory Animals include the following

### **Rabbit (*Oryctolagus cuniculus*)**

Uses:

- (1) Pathogenic *Treponema* does not grow in artificial culture media, and it maintained for many decades by serial testicular passage in rabbits.
- (2) It differentiate between human and bovine variety of tuberculosis, where animal is susceptible to later.
- (3) It is used to prepare the diagnostic sera.
- (4) It also useful in virus isolation and identification.

### **Guinea Pig (*Cavia porcelus*)**

Uses: It is used: (1) To isolate human and bovine variety of tuberculosis and animal is susceptible to both. (2) To do virulence test in diphtheria and tetanus. (3) To obtain the complement for CFT. (4) To investigate the typhus fever. (5) To investigate the Enteric invasive *Escherichia coli* (EIEC) or *Shigella* called Sereny test or *Listeria* called Anton test. (6) To isolate and to identify the viruses. (7) To diagnose the anaphylaxis. (8) It was used to establish many theories like germ theory by Pasteur, Koch's phenomenon, Pfiffer's phenomenon and Theobald Smith phenomenon (anaphylaxis)

### **Rat**

Black rat: *Rattus rattus*.

Brown rat: *Rattus norvegicus*.

White rat or albino rat: *Rattus albus*.

Identification features: These are larger than mouse, rough fur and different colors as mentioned above.

Uses:

- (1) It is useful to differentiate between *Yersinia pseudotuberculosis* and *Yersinia pestis*.
- (2) It also associated with other zoonotic pathogens like *Leptospira*, *T. gondii*, *C. jejuni*, etc.

### **Mouse (*Mus musculus*)**

Identification features: It is smaller than rat and has smooth fur.

Uses:

- (1) It is very susceptible to pneumococcus and useful to isolate it from the infective materials.
- (2) Foot pad of mouse is useful to cultivate the *M. leprae*.
- (3) It is used to demonstrate the ascending and descending tetanus. (4) It is used to demonstrate the influenza virus and Chlamydia.
- (4) Infant suckling mouse, but not the adult mouse is useful in virus isolation and identification like coxsackie viruses.
- (5) It is used to isolate the causative organisms of relapsing fever, rat bite fever and trypanosomiasis.

### **Sheep**

Uses:

- (1) Sheep blood is useful to prepare the blood agar.
- (2) RBCs of sheep are useful in CFT, Paul Bunnell test, SRBC (sheep red blood cells) rosette or E (erythrocyte) rosette formation with CD2 cells and EAC (erythrocyte antibody complement) rosette formation with CR2 (complement receptor-2).

### **Monkey**

Use: Monkey kidney cells called Vero cells are useful for viral culture.